

Chapter 2 Learning content

Learning Content of Junior Secondary Mathematics Curriculum

Notes:

1. *Learning units are grouped under three strands (“Number and Algebra”, “Measures, Shape and Space” and “Data Handling”) and a Further Learning Unit.*
2. *Related learning objectives are grouped under the same learning unit.*
3. *The learning objectives underlined are the Non-foundation Topics and the learning objectives with asterisks (**) are the Enrichment Topics.*
4. *The notes in the “Remarks” column of the table may be considered as supplementary information about the learning objectives.*
5. *To aid teachers in judging how far to take a given topic, a suggested lesson time in hours is given against each learning unit. However, the lesson time assigned is for their reference only. Teachers may adjust the lesson time to meet their individual needs.*
6. *The total lesson time for junior secondary Mathematics curriculum is 331 – 413 hours (i.e. 12% – 15% of the total lesson time available for the junior secondary curriculum).*

Learning Unit	Learning Objective	Time	Remarks
Number and Algebra Strand			
1. Basic computation	1.1 recognise the tests of divisibility of 4, 6, 8 and 9	8	At Key Stage 2, students are required to recognise the tests of divisibility of 2, 3, 5 and 10.

Learning Unit	Learning Objective	Time	Remarks
	1.2 understand the concept of power 1.3 perform prime factorisation of positive integers 1.4 find the greatest common divisor and the least common multiple 1.5 perform mixed arithmetic operations of positive integers involving multiple levels of brackets		Computations involving powers are not required. Students are required to use prime factorisation and short division to find the greatest common divisor and the least common multiple. At Key Stage 2, students are required to find the greatest common divisor and the least common multiple of two numbers by listing their multiples and factors, and using short division. The terms “H.C.F.”, “gcd”, etc. can be used.

Learning Unit	Learning Objective	Time	Remarks
	1.6 perform mixed arithmetic operations of fractions and decimals		At Key Stage 2, students are required to perform mixed arithmetic operations of three numbers (including fractions and integers). In the mixed operations of addition and subtraction involving three fractions with different denominators, all denominators should not exceed 12. The above restrictions are no longer applicable at Key Stage 3, but complicated operations should be avoided. Note: Teachers are suggested to arrange this learning unit as the first one to be taught at this key stage.
2. Directed numbers	2.1 understand the concept of directed numbers	9	Students are required to represent the directed numbers on the number line. Students are required to compare the magnitude of directed numbers.

Learning Unit	Learning Objective	Time	Remarks
	2.2 perform mixed arithmetic operations of directed numbers 2.3 solve problems involving directed numbers		
3. Approximate values and numerical estimation	3.1 recognise the concept of approximate values 3.2 understand the estimation strategies 3.3 solve related real-life problems	6	Students are required to round off a number to a certain number of significant figures, a certain place and a certain number of decimal places. At Key Stage 2, students are required to round off a whole number to a certain place and round off a decimal to the nearest tenths or hundredths. The estimation strategies include rounding off, rounding up and rounding down.

Learning Unit	Learning Objective	Time	Remarks
	3.4 **design numerical estimation strategies according to the contexts and judge the reasonableness of the results obtained		
4. Rational and irrational numbers	4.1 recognise the concept of nth root 4.2 recognise the concepts of rational and irrational numbers 4.3 <u>perform mixed arithmetic operations of simple quadratic surds $a\sqrt{b}$</u>	7	Computations involving nth roots are not required. Students are required to evaluate expressions such as $\sqrt[3]{-8}$. Students are required to represent rational and irrational numbers on the number line. In the simple quadratic surds $a\sqrt{b}$, a is a rational number, b is a positive rational number and $a\sqrt{b}$ is an irrational number. The computations such as the following examples are required:

Learning Unit	Learning Objective	Time	Remarks
	4.4 **explore the relation between constructible numbers and rational and irrational numbers		• $\sqrt{3} + \sqrt{12} = 3\sqrt{3}$ • $\frac{8}{3\sqrt{2}} = \frac{4\sqrt{2}}{3}$ The complicated mixed arithmetic operations are not required, for example, $\frac{1}{2 + \sqrt{3}} = 2 - \sqrt{3}$.
5. Using percentages	5.1 understand the concept of percentage changes	15	Percentage increase and percentage decrease are required. Percentage change can also be called “percentage of change”.

Learning Unit	Learning Objective	Time	Remarks
	5.2 solve related real-life problems		The problems include those about discount and profit or loss, growth and depreciation, simple and compound interests, successive and component changes, and salaries tax.
6. Rates, ratios and proportions	6.1 understand the concepts of rates, ratios and proportions 6.2 solve problems involving rates, ratios and proportions	8	Direct and inverse proportions are required. Students are required to solve problems about plans involving scales. Problems involving similar figures are tackled in Learning Objectives 18.3 and 22.3. Teachers may consider using real-life examples or related learning elements in Science Education or Technology Education KLAs to enhance learning and teaching.

Learning Unit	Learning Objective	Time	Remarks
7. Algebraic expressions	7.1 represent word phrases by algebraic expressions	7	Notations such as ab representing $a \times b$, $\frac{a}{b}$ representing $a \div b$, are required. Students are required to use algebraic expressions to represent formulae, for example, the formula of the area of a triangle $A = \frac{bh}{2}$ that was learnt at Key Stage 2.
	7.2 represent algebraic expressions by word phrases		
	7.3 recognise the concept of sequences of numbers		Students are required to guess the next term of a sequence and give explanations. Students are required to find a particular term from the general term of a sequence. Sequences of odd numbers, even numbers, square numbers, and triangular numbers are required.

Learning Unit	Learning Objective	Time	Remarks
	7.4 recognise the preliminary idea of functions		The concept of input-processing-output is required. Note: The algebraic expressions discussed in this Learning Unit are confined to expressions involving addition, subtraction, multiplication, division and powers of numbers or variables.
8. Linear equations in one unknown	8.1 solve linear equations in one unknown 8.2 formulate linear equations in one unknown from a problem situation 8.3 solve problems involving linear equations in one unknown	7	
9. Linear equations in two unknowns	9.1 understand the concept of linear equations in two unknowns and their graphs	12	Students should understand: the graph of a linear equation in two

Learning Unit	Learning Objective	Time	Remarks
			unknowns is a straight line - the coordinates of every point lying on a straight line satisfy the corresponding linear equation in two unknowns - the coordinates of every point not lying on a straight line do not satisfy the corresponding linear equation in two unknowns
	9.2 solve simultaneous linear equations in two unknowns by the graphical method		Students are required to recognise that the exact values of the solutions may not be obtained by the graphical method.
	9.3 solve simultaneous linear equations in two unknowns by the algebraic methods		The algebraic methods include substitution and elimination. The simultaneous equations include those with: no solutions

Learning Unit	Learning Objective	Time	Remarks
	9.4 formulate simultaneous linear equations in two unknowns from a problem situation 9.5 solve problems involving simultaneous linear equations in two unknowns		- only one solution - more than one solution
10. Laws of integral indices	10.1 understand the laws of positive integral indices	11	The laws of positive integral indices include: $a^p a^q = a^{p+q}$ $\frac{a^p}{a^q} = a^{p-q}$ $(a^p)^q = a^{pq}$ $a^p b^p = (ab)^p$ $\frac{a^p}{b^p} = \left(\frac{a}{b}\right)^p$

Learning Unit	Learning Objective	Time	Remarks
	10.2 understand the definitions of zero exponent and negative exponents 10.3 understand the laws of integral indices 10.4 understand scientific notations 10.5 <u>understand the binary number system and the denary number system</u> 10.6 **understand other numeral systems, such as the hexadecimal number system		The above laws of positive integral indices also apply to integral indices. Teachers may consider using real-life examples or related learning elements in Science Education or Technology Education KLAs to enhance learning and teaching. Interconversion between binary numbers and denary numbers is required. Teachers may consider using real-life examples or related learning elements in Science Education or Technology Education KLAs to enhance learning and teaching.

Learning Unit	Learning Objective	Time	Remarks
11. Polynomials	11.1 understand the concept of polynomials 11.2 perform addition, subtraction, multiplication and their mixed operations of polynomials 11.3 factorise polynomials	15	Students are required to understand the concepts of terms, monomials, binomials, orders, powers, constant terms, like terms, unlike terms and coefficients. Students are required to arrange the terms of a polynomial in ascending order or descending order. The operations of polynomials involving more than one variable are required. Students are required to understand that factorisation can be regarded as a reverse process of expansion of polynomials. The following methods are required: • extracting common factors (and grouping of terms) • cross-method

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12. Identities	12.1 understand the concept of identities 12.2 use identities to expand algebraic expressions 12.3 use identities to factorise polynomials	8	Students are required to understand the differences between equations and identities, and to prove identities. The identities include: • Difference of two squares $(a - b)(a + b) \equiv a^2 - b^2$ • Perfect square $(a \pm b)^2 \equiv a^2 \pm 2ab + b^2$ The identities include: • Difference of two squares $a^2 - b^2 \equiv (a - b)(a + b)$ • Perfect square $a^2 \pm 2ab + b^2 \equiv (a \pm b)^2$

Learning Unit	Learning Objective	Time	Remarks
13. Formulae	13.1 perform operations of algebraic fractions 13.2 use substitution to find the values of unknowns in the formulae 13.3 change the subject of formulae not involving radical signs	9	Algebraic fractions are confined to those having denominators as products of linear factors. Note: Teachers may consider using real-life examples or related learning elements in Science Education or Technology Education KLAs to enhance learning and teaching.
14. Linear inequalities in one unknown	14.1 understand the concept of inequalities	6	The following are required: • use inequalities to represent word phrases • represent the following inequalities on the number line: $x > a$, $x \geq a$, $x < a$, $x \leq a$

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	14.2 recognise the basic properties of inequalities		The properties include • if $a > b$ and $b > c$, then $a > c$ • if $a > b$, then $a \pm c > b \pm c$ • if $a > b$ and c is positive, then $ac > bc$ and $\frac{a}{c} > \frac{b}{c}$ • if $a > b$ and c is negative, then $ac < bc$ and $\frac{a}{c} < \frac{b}{c}$ where “>” and “<” in the above properties can be replaced by “$\geq$” and “$\leq$” respectively.
	14.3 solve linear inequalities in one unknown		Students are required to represent solutions of inequalities on the number line.
	14.4 solve problems involving linear inequalities in one unknown		

Learning Unit	Learning Objective	Time	Remarks
Measures, Shape and Space Strand			
15. Errors in measurement	15.1 recognise the concept of errors in measurement 15.2 recognise the concepts of maximum absolute errors, relative errors and percentage errors 15.3 solve problems related to errors 15.4 **design estimation strategies in measurement according to the contexts and judge the reasonableness of the results obtained	6	
16. Arc lengths and areas of sectors	16.1 understand the formula for arc lengths of circles 16.2 understand the formula for areas of sectors of circles	8	Students are required to understand the property that the arcs are proportional to their corresponding angles at the centre. “Find the diameter or radius of a circle from its area” is not required in the primary Mathematics curriculum.

Learning Unit	Learning Objective	Time	Remarks
	16.3 solve problems related to arc lengths and areas of sectors of circles 16.4 **recognise the Circle Dissection Algorithm of the ancient Chinese mathematician Liu Hui and further recognise Huilu and Tsulu (approximations of π)		The problems on finding perimeters and areas of composite figures are required.
17. 3-D figures	17.1 recognise the concepts of right prisms, right circular cylinders, right pyramids, right circular cones, regular prisms, regular pyramids, polyhedra and spheres 17.2 recognise the sections of prisms, circular cylinders, pyramids, circular cones, polyhedra and spheres	5	Students are required to recognise the concept of regular tetrahedra. Students are required to recognise: - different sections of the 3-D figure can have different sizes and shapes - the concept of uniform cross sections At Key Stage 2, students are required to recognise that the sizes and shapes of the

Learning Unit	Learning Objective	Time	Remarks
	17.3 sketch the 2-D representations of 3-D figures 17.4 **recognise the three orthographic views of 3-D figures 17.5 **recognise Euler's formula and explore the number of regular polyhedra (Platonic solids)		cross sections of prisms and cylinders are the same as that of the bases if they are parallel to the bases, but the term “uniform cross sections” is not introduced. Students are required to sketch the 2-D representations of right prisms, right circular cylinders, right pyramids and right circular cones. Students may use the tools such as oblique grids and isometric grids to learn the sketching of 2-D representations. Three orthographic views of 3-D figures are not required.

Learning Unit	Learning Objective	Time	Remarks
18. Mensuration	18.1 recognise the formulae for volumes of prisms, circular cylinders, pyramids, circular cones and spheres 18.2 find the surface areas of right prisms, right circular cylinders, right pyramids, right circular cones and spheres 18.3 recognise the relations among lengths, areas and volumes of similar figures and solve related problems 18.4 solve problems involving volumes and surface areas	15	Students are required to recognise the projection of a point on a plane and the concept of height of a 3-D figure. Students are required to recognise the formula for surface areas of spheres. Students are required to recognise the concept of similar 3-D figures. The concept of similar 2-D figures is dealt with in Learning Objective 22.3. Students are required to recognise frustums and solve problems related to their surface areas and volumes.

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	18.5 **explore ways to form a container with the greatest capacity by folding an A4-sized paper with squares cutting from its four corners		Note: Students are required to understand 2-D representations of 3-D figures.
19. Angles and parallel lines	19.1 understand the concepts and properties of adjacent angles on a straight line, vertically opposite angles and angles at a point	11	The properties include: • the sum of adjacent angles on a straight line is equal to a straight angle • vertically opposite angles are equal • the sum of angles at a point is equal to a round angle Students are required to recognise the concepts of complementary angles and supplementary angles.

Learning Unit	Learning Objective	Time	Remarks
	19.2 understand the concepts of corresponding angles, alternate interior angles and interior angles		Students are required to recognise transversals.
	19.3 recognise the conditions for two straight lines being parallel		The conditions include: • alternate interior angles are equal • corresponding angles are equal • interior angles are supplementary
	19.4 recognise the angle properties associated with parallel lines		The properties include: • alternate interior angles of parallel lines are equal • corresponding angles of parallel lines are equal • interior angles of parallel lines are supplementary
	19.5 understand the properties of the interior and		The properties include:

Learning Unit	Learning Objective	Time	Remarks
	exterior angles of triangles		- the angle sum of a triangle is equal to a straight angle - the exterior angle of a triangle is equal to the sum of the interior opposite angles of the triangle Students are required to recognise the concepts of acute-angled triangles and obtuse-angled triangles.
20. Polygons	20.1 understand the concept of regular polygons 20.2 understand the formula for the sum of the interior angles of a polygon 20.3 understand the formula for the sum of the exterior angles of a convex polygon 20.4 <u>appreciate the triangles, quadrilaterals, and regular polygons that tessellate in the plane</u>	8	

Learning Unit	Learning Objective	Time	Remarks
	20.5 <u>construct equilateral triangles and regular hexagons with compasses and straightedge</u> 20.6 <u>**explore ways to construct regular pentagons with compasses and straightedge</u>		Teachers may let students recognise the basic knowledge of construction with compasses and straightedge. Students may use information technology for construction.
21. Congruent triangles	21.1 understand the concept of congruent triangles 21.2 recognise the conditions for congruent triangles 21.3 understand the property of isosceles triangles	14	The conditions include: SAS, SSS, ASA, AAS and RHS. The property refers to: the base angles of an isosceles triangle are equal. Teachers may allow students to recognise the proof of equal base angles in isosceles triangles by means of SAS.

Learning Unit	Learning Objective	Time	Remarks
	21.4 understand the condition for isosceles triangles 21.5 <u>construct angle bisectors, perpendicular bisectors, perpendicular lines, parallel lines, special angles and squares with compasses and straightedge</u> 21.6 <u>recognise the concept of congruent 2-D figures</u> 21.7 <u>**explore the angles that can be constructed with compasses and straightedge</u>		The condition refers to: if two angles of a triangle are equal, then the triangle is isosceles. Students may use information technology for construction.
22. Similar triangles	22.1 understand the concept of similar triangles 22.2 recognise the conditions for similar triangles	9	The conditions include : • AAA (AA) • corresponding sides are proportional • two corresponding sides proportional and their included angles equal

Learning Unit	Learning Objective	Time	Remarks
	22.3 recognise the concept of similar 2-D figures 22.4 **explore shapes of fractals		Students are required to recognise that quadrilaterals with corresponding sides proportional are not necessarily similar.
23. Quadrilaterals	23.1 understand the properties of parallelograms 23.2 understand the properties of rectangles, rhombuses and squares	13	The properties include: • opposite sides equal, opposite angles equal and diagonals bisect each other The properties of rectangles include: • all the properties of parallelograms • the two diagonals equal • the diagonals bisect each other into four equal line segments The properties of rhombuses include: • all the properties of parallelograms

Learning Unit	Learning Objective	Time	Remarks
	23.3 <u>understand the conditions for parallelograms</u>		- the diagonals are perpendicular to each other - the diagonals bisect the opposite angles The properties of squares include: - all the properties of rectangles - all the properties of rhombuses - the diagonals form a 45° with the sides of the squares The conditions include: - opposite sides are equal - opposite angles are equal - diagonals bisect each other - there is a pair of sides that are both equal and parallel

Learning Unit	Learning Objective	Time	Remarks
	23.4 <u>use the above properties or conditions to perform simple geometric proofs</u> 23.5 <u>understand the mid-point theorem and the intercept theorem</u> 23.6 **explore the conditions for congruent quadrilaterals		Note: - Students are required to recognise logical relations such as “squares are figures having four equal sides, but figures having four equal sides are not necessarily squares”. - Teachers may let students recognise the deductive method in Euclid’s <i>Elements</i>.

Learning Unit	Learning Objective	Time	Remarks
24. Centres of triangles	24.1 understand the properties of angle bisectors and perpendicular bisectors 24.2 <u>understand the concurrence of angle bisectors and the concurrence of perpendicular bisectors of a triangle</u>	8	The properties include: • if a point lies on the angle bisector, then it is equidistant from the two sides of the angle, and the converse • if a point lies on the perpendicular bisector of a line segment, then it is equidistant from the two end points of the line segment, and the converse Students are required to recognise the concepts of the incentre and the circumcentre of a triangle, and the following properties: • the incentre of a triangle is equidistant from the three sides of the triangle, and a circle inside the triangle can be constructed with this distance as its radius and the incentre as its centre

Learning Unit	Learning Objective	Time	Remarks
	24.3 <u>recognise the concurrence of medians and the concurrence of altitudes of a triangle</u>		the circumcentre of a triangle is equidistant from the three vertices of the triangle and a circle passing through the vertices can be constructed with this distance as its radius and the circumcentre as its centre Students are required to recognise the concepts of the centroid and the orthocentre of a triangle. Teachers may use information technology to help students understand the proofs of concurrence of medians and concurrence of altitudes.
25. Pythagoras' theorem	25.1 understand the Pythagoras' theorem	6	Teachers may introduce different proofs of Pythagoras' theorem, for example, the proof by the ancient Chinese mathematician Liu Hui and the proof in Euclid's <i>Elements</i> .

Learning Unit	Learning Objective	Time	Remarks
	25.2 recognise the converse of Pythagoras' theorem 25.3 solve problems related to Pythagoras' theorem and its converse 25.4 **explore Pythagorean triples		Teachers may introduce the Pythagorean school and its related history, including the history of the first mathematical crisis. Teachers may introduce the proofs of the converse of Pythagoras' theorem.
26. Rectangular coordinate system	26.1 recognise the rectangular coordinate system 26.2 find the distance between two points on a horizontal line and the distance between two points on a vertical line	19	Students are required to • represent the position of a point by its coordinates • mark the point with given coordinates

Learning Unit	Learning Objective	Time	Remarks
	26.3 find areas of polygons in the rectangular coordinate plane 26.4 recognise the effect of transformations on a point in the rectangular coordinate plane 26.5 understand the distance formula 26.6 understand the mid-point formula and <u>the formula for the internal point of division</u> 26.7 understand the slope formula and solve related problems		Transformations include: • translation • reflection in a line parallel to the x-axis or y-axis • clockwise or anti-clockwise rotation about the origin through $n \cdot 90^\circ$, where n is a positive integer Students are required to recognise the concept of intercepts.

Learning Unit	Learning Objective	Time	Remarks
	26.8 recognise the relation between the slopes of parallel lines and the relation between the slopes of perpendicular lines, and solve related problems 26.9 use coordinate geometry to perform simple geometric proofs		Besides finding slopes, students are required to use the slope formula to find x-coordinates or y-coordinates of points on straight lines, x-intercepts or y-intercepts of straight lines, from given conditions, for example: given the coordinates of the two points on a straight line, find the x-intercept or y-intercept of the straight line Students are required to identify parallel lines and perpendicular lines from their slopes. The relation between slope and inclination in the rectangular coordinate plane is dealt with in the Compulsory Part at Key Stage 4.

Learning Unit	Learning Objective	Time	Remarks
	26.10 **explore the formula for the external point of division		
27. Trigonometry	27.1 understand sine, cosine and tangent of angles between 0° and 90° 27.2 understand the properties of trigonometric ratios	18	The trigonometric ratios of 0° and 90° are not required. The properties include: For $0^\circ < \theta < 90^\circ$, • as θ increases, the values of $\sin \theta$ and $\tan \theta$ increase and that of $\cos \theta$ decreases • $0 < \sin \theta < 1$ • $0 < \cos \theta < 1$ • $\tan \theta > 0$ • $\frac{\sin \theta}{\cos \theta} = \tan \theta$

Learning Unit	Learning Objective	Time	Remarks
	27.3 understand the exact values of trigonometric ratios of 30°, 45° and 60° 27.4 solve problems related to plane figures 27.5 solve problems involving gradients, angles of elevation, angles of depression and bearings		• $\sin^2 \theta + \cos^2 \theta = 1$ • $\sin(90^\circ - \theta) = \cos \theta$ • $\cos(90^\circ - \theta) = \sin \theta$ • $\tan(90^\circ - \theta) = \frac{1}{\tan \theta}$ Students are required to recognise the relation between gradients and inclinations. Students are required to recognise two kinds of bearing such as 010° and $N10^\circ E$.

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Data Handling Strand			
28. Organisation of data	28.1 recognise the concepts of discrete data and continuous data 28.2 recognise organisation of data without grouping 28.3 recognise organisation of data in groups	4	Note: Students are required to recognise the organisation of data using frequency distribution tables.
29. Presentation of data	29.1 recognise stem-and-leaf diagrams and histograms	17	Students are required to construct simple stem-and-leaf diagrams and histograms using paper and pen. When constructing stem-and-leaf diagrams and histograms of a larger amount of data, students may use information technology. Students are required to recognise the

Learning Unit	Learning Objective	Time	Remarks
	29.2 interpret stem-and-leaf diagrams and histograms 29.3 interpret statistical charts representing two different sets of data in daily life 29.4 recognise frequency polygons, frequency curves, cumulative frequency polygons and cumulative frequency curves 29.5 interpret frequency polygons, frequency curves, cumulative frequency polygons and cumulative frequency curves		construction of statistical charts in appropriate scales. Example: Temperature and rainfall charts Construction of statistical charts is required. Students are required to find the following from cumulative frequency polygons and cumulative frequency curves: • medians, quartiles (upper quartiles, lower quartiles) and percentiles • the positions of individual data in the populations

Learning Unit	Learning Objective	Time	Remarks
	29.6 choose appropriate statistical charts to present data 29.7 recognise the uses and abuses of statistical charts in daily life		Statistical charts include stem-and-leaf diagrams and histograms, and those that are dealt with at Key Stage 2, including bar charts, pie charts and broken line graphs. Teachers may consider using real-life examples or related learning elements in Science Education or Technology Education KLAs to enhance learning and teaching.
30. Measures of central tendency	30.1 understand the concepts of mean, median and mode/modal class	10	Students are required to understand the features and limitations of each measure, for example, a single extreme datum may have a great influence on the mean, and the median is not affected by a single extreme datum. Mean can also be called “average”. At Key Stage 2, students are required to recognise the concept of average.

Learning Unit	Learning Objective	Time	Remarks
	30.2 calculate mean, median and mode of ungrouped data 30.3 calculate mean, median and modal class of grouped data 30.4 recognise the uses and abuses of mean, median and mode/modal class in daily life 30.5 <u>understand the effects of the following operations on the mean, median, and mode:</u> (i) <u>adding a common constant to each item of the set of data</u> (ii) <u>multiplying each item of the set of data by a common constant</u>		Students are required to understand that mean and median of grouped data are estimations only.

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	30.6 recognise the concept of weighted mean 30.7 solve problems involving weighted mean		Real-life applications are required, for example: the methods for calculating average scores for report cards and scores for university admissions.
31. Probability	31.1 recognise the concepts of certain events, impossible events and random events 31.2 recognise the concept of probability 31.3 calculate probabilities of events by listing the sample space and counting 31.4 solve problems involving probability	12	Geometric probability is not required. Students may use diagrams such as Venn diagrams to understand the concept of sample space. Students are required to use tables or tree diagrams to list sample spaces.

Learning Unit	Learning Objective	Time	Remarks
	31.5 <u>recognise the concept of expectation</u> 31.6 <u>solve problems involving expectation</u>		
Further Learning Unit			
32. Inquiry and investigation	Through various learning activities, discover and construct knowledge, further improve the ability to inquire, communicate, reason and conceptualise mathematical concepts	20	This is not an independent and isolated learning unit. The time is allocated for students to engage in learning activities from different learning units, for example, activities on enrichment topics, cross-learning unit activities, and cross-KLA activities that based on mathematical topics.

Total lesson time: 331 hours